The chain

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06 Deposition and Etch
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Chapter 14 Stage 14 of 15 High

Systems and Networking

Nvidia's and AMD's racks wire 72 AI chips together to run as one computer. Many firms can assemble the racks, but lasers for the optical links were scarce in 2026.

Specimen 14 · Systems and Networking

An eight-GPU baseboard, the board that carries eight accelerator modules and the switch chips joining them. It is about 510 by 400 mm and drawn to scale, with the module frames standing about 30 mm above the board. The two nickel-gray squares in the middle are the NVLink switch chips.

In plain terms

One AI chip is not enough to train a model. So seventy-two of them are wired together in a rack, a cabinet the size of a wardrobe, and run as one computer. Many such racks are then linked to train a large model. Each chip sits on its own board and draws more than a thousand amps, several times the current a whole house is wired to carry. The chips have to swap results with each other all the time. A copper wire can carry those signals only a few meters before they fade, so the longer links send them as light down glass fiber. Many companies can build the rack, but in 2026 the lasers that send the light were scarce.

In short

In March 2026 Nvidia agreed to invest $2 billion in the laser maker Lumentum, alongside a multibillion-dollar purchase commitment and rights to future capacity for advanced laser components[3]. Indium phosphide is used in high-speed lasers for optical communications, and China produced 70 percent of the world's indium in 2025[13].

On 4 February 2025 China added indium phosphide substrates, the wafers the lasers are grown on, to its export control list[14]. AXT, a US-listed maker of those wafers, does all its manufacturing in China[14]. In its filing for the quarter to June 2026, AXT said it could not estimate when it would resume shipping the wafers to the United States[14].

3 NVIDIA Announces Strategic Partnership With Lumentum to Develop... Nvidia, 2 March 2026. nvidianews.nvidia.com
13 Mineral Commodity Summaries 2026: Indium. U.S. Geological Survey, 5 February 2026. pubs.usgs.gov
14 AXT INC, Form 10-Q quarterly report for the period ended 2026-06-30. U.S. Securities and Exchange Commission, 13 August 2026. sec.gov

Chokepoint card

Systems and Networking, stage 14 of 15

Concentration

High

Four to six firms could supply this stage today, and the top two take most of the revenue.

Substitutability

Moderate

Rated on NVLink and its NVSwitch chips, which Nvidia controls and which join the accelerators in a rack[1]. Separately, lasers for the optical links were scarce in 2026[2]. Nvidia secured future capacity rights for them from Lumentum in March 2026[3].

Price or market size

$3.1 million

for one GB200 NVL72 rack bought by a typical hyperscaler, or about $3.9 million with networking, storage and other items, by SemiAnalysis's August 2025 estimate[4]

Who leads	US	**Nvidia**	NVLink, InfiniBand and Spectrum-X Ethernet, which it credited for data center growth in the quarter to April 2026[5]
	US	**Broadcom**	$16.7B of AI semiconductor revenue in the quarter to August 2026, up 221%[6]
	TW	**Foxconn**	World's largest AI server provider, its chairman said in March 2026[7]
	CN	**InnoLight**	High-speed optical modules made in Suzhou and, since 2019, Thailand[8]
	US	**Coherent**	Lasers and optical networking products, with a $2B Nvidia investment agreed in March 2026[9]
Where it is made	US	**United States**	Switch and accelerator silicon, system design, indium phosphide lasers
	TW	**Taiwan**	Contract design, rack assembly and high-layer-count circuit boards
	JP	**Japan**	Low-loss board laminate
	CN	**China**	Optical module assembly and test, and most of the world's indium
	TH	**Thailand**	Chinese module makers' offshore transceiver plants
Why substitution is possible			UALink is an open specification for the same job as NVLink[10]. AMD's Helios rack, launched in July 2026, runs it over Ethernet[11]. Since May 2025 Nvidia's NVLink Fusion program has let other firms' custom chips use NVLink, which keeps those firms tied to Nvidia's interconnect[12].

1 NVIDIA CORP, Form 10-K annual report for the period ended 2026-01-25 (10-K). U.S. Securities and Exchange Commission, 25 February 2026. sec.gov

2 AI demand strains supplies of lasers, fiber and other optical tech. Nikkei Asia, 27 May 2026. asia.nikkei.com

3 NVIDIA Announces Strategic Partnership With Lumentum to Develop... Nvidia, 2 March 2026. nvidianews.nvidia.com

4 H100 vs GB200 NVL72 Training Benchmarks - Power, TCO, and Reliability Analysis... SemiAnalysis, 20 August 2025. newsletter.semianalysis.com

5 NVIDIA CORP, Form 10-Q quarterly report for the period ended 2026-04-26 (10-Q). U.S. Securities and Exchange Commission, 20 May 2026. sec.gov

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7 Hon Hai Technology Group (Foxconn) Announces FY2025 & 4Q25 Financial Results. Hon Hai Precision Industry (Foxconn), 16 March 2026. honhai.com

8 InnoLight Technology, company overview page. InnoLight Technology, accessed 9 September 2026. innolight.com

9 NVIDIA and Coherent Announce Strategic Partnership to Develop Optics Technology... Nvidia, 2 March 2026. nvidianews.nvidia.com

10 UALink™ 200G 1.0 Specification Overview. UALink Consortium, 15 May 2025. ualinkconsortium.org

11 AAI 2026: AMD Launches AMD Helios Rackscale Solution for Frontier AI. AMD Newsroom, 23 July 2026. newsroom.amd.com

12 NVIDIA Unveils NVLink Fusion for Industry to Build Semi-Custom AI... Nvidia, 18 May 2025. nvidianews.nvidia.com

Where China stands

Chinese firms such as InnoLight assemble high-speed optical modules in volume, at sites including Suzhou and, since 2019, Thailand[8]. In February 2025 China put indium under new export restrictions[13]. The same month it began requiring export permits for indium phosphide substrates, the wafers that lasers for optical links are grown on[14].

Where the US stands

Coherent broke ground in Sherman, Texas, in June 2026 on an expansion of its factory there, which makes optical devices from indium phosphide[15]. In August 2026 Nvidia said that Wistron's new plant in Fort Worth, Texas, was producing its GB300 superchips and that Foxconn was building a factory in Houston[16].

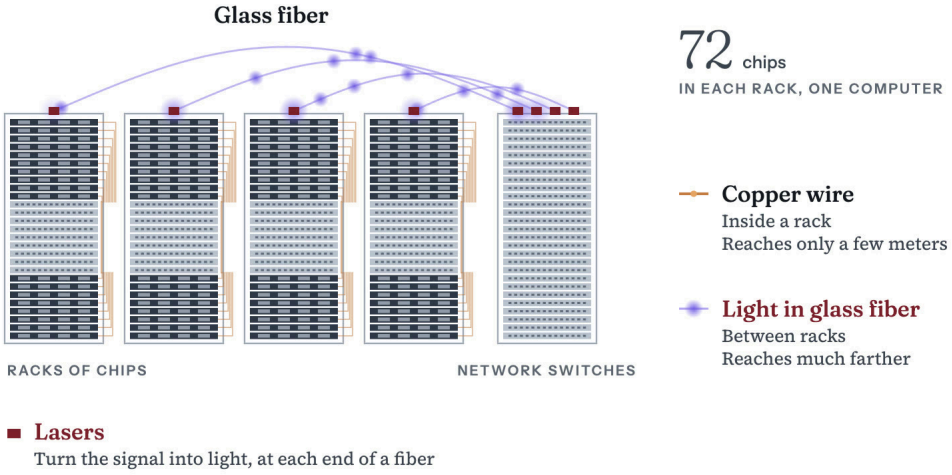
Before a chip package can compute, it has to be built into a machine. Building that machine is a separate industry from making the package.

01

How it works

Figure 14.1 · Explainer, 20 seconds

How AI chips are wired together



1. **One rack.** One AI chip is not enough to train a model. So 72 of them are wired together in one rack, a cabinet the size of a wardrobe, and run as one computer.
2. **Copper inside.** Inside the rack, copper wires link each tray of chips to the switch trays, which pass data between any two chips. Copper carries a signal only a few meters.
3. **Light between.** Training a large model takes many racks, most of them too far apart for copper. The links between them carry the signal as flashes of light in glass fiber.
4. **Lasers.** At each end of a fiber, a laser turns the electrical signal into flashes of light, and a detector turns them back. That lets thousands of chips in many racks work on one model.

Note Simplified. The trays follow the layout of an Nvidia NVL72 rack. Racks and fibers are not drawn to scale.

An accelerator (the AI chip) ships as a module, the chip package on a small board with the circuits that feed it power. Nvidia calls its version SXM, and the open-standards equivalent is the Open Compute Project's accelerator

8 InnoLight Technology, company overview page. InnoLight Technology, accessed 9 September 2026. innolight.com
13 Mineral Commodity Summaries 2026: Indium. U.S. Geological Survey, 5 February 2026. pubs.usgs.gov
14 AXT INC, Form 10-Q quarterly report for the period ended 2026-06-30. U.S. Securities and Exchange Commission, 13 August 2026. sec.gov
15 Coherent Announces a CHIPS Letter of Intent for $50 Million to Expand... Coherent, 16 June 2026. coherent.com
16 NVIDIA and Partners Build in America, for America. Nvidia, 5 August 2026. blogs.nvidia.com

module. Eight modules bolt onto a baseboard that carries their power and the links between them. On Nvidia's baseboard, switch chips pass data between any two modules at full speed.

Rack-scale systems drop the baseboard and put the chips on trays stacked in the rack. An Nvidia GB300 NVL72 holds 18 compute trays plus nine switch trays, 72 GPUs in all, and draws up to 142 kW[17]. The newer Vera Rubin NVL72 uses the same rack footprint[18]. The research firm SemiAnalysis estimated in August 2025 that a rack of the earlier GB200 NVL72 model cost a typical hyperscaler $3.1 million, or about $3.9 million with networking, storage and other items[4].

Inside the rack the trays reach each other over copper[17]. Between neighboring racks, the signal can travel up to 7 meters over active electrical cables, a type of copper cable sold by firms such as Credo[19]. Longer links run on glass fiber, with an optical transceiver at each end. A laser in the sending transceiver turns the electrical signal into pulses of light, and a detector in the receiving one turns the pulses back into current.

The accelerators in an AI cluster exchange data over two networks that are not interchangeable. Scale-up connects anywhere from a few dozen accelerators to thousands closely enough that they can read each other's memory. It runs at terabytes per second. Scale-out joins those groups to one another at hundreds of gigabits per second. Scale-up speeds are quoted in bytes and scale-out speeds in bits, and a byte is eight bits. Each GPU in a GB300 NVL72 can send 900 gigabytes per second each way over the scale-up network, or 7,200 gigabits per second[20]. Over the scale-out network it can send 800 gigabits per second[20]. So each GPU has nine times as much bandwidth for scale-up as for scale-out, SemiAnalysis calculated in January 2026[20].

Scale-up bandwidth limits how efficiently a large model trains. A model too big for one accelerator is split across many. The parts that exchange data constantly stay inside the scale-up group, because they need its high bandwidth[21]. Whole blocks of layers, the stages a model's calculation passes through in turn, can sit on different groups across the slower scale-out network[21].

02

Variants and trade-offs

Scale-up: NVLink, UALink and Google's torus

The designs differ in how many accelerators can share one pool of memory.

- **NVLink and NVSwitch.** Nvidia's NVLink bandwidth per GPU, counting both directions, went from 900 GB/s in NVLink 4 to 1,800 GB/s in NVLink

4 H100 vs GB200 NVL72 Training Benchmarks - Power, TCO, and Reliability Analysis... SemiAnalysis, 20 August 2025. newsletter.semianalysis.com

17 System Hardware & Components. Nvidia, accessed 9 September 2026. docs.nvidia.com

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21 Parallelisms Guide — Megatron Bridge. Nvidia, accessed 9 September 2026. docs.nvidia.com

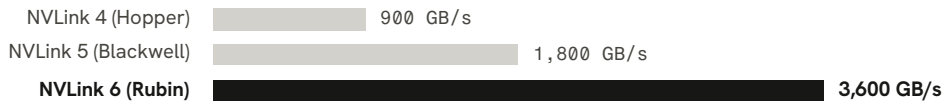
5[22]. Across a whole Blackwell NVL72 rack, NVLink moves 130 TB/s[22]. Nvidia's Rubin launch release of January 2026 gives NVLink 6 as 3,600 GB/s per GPU and 260 TB/s for a Vera Rubin NVL72 rack[23]. As of October 2026, Nvidia's NVLink product page still lists preliminary figures of 3,000 GB/s and 216 TB/s[22].

- **UALink.** The open specification, from the UALink Consortium, runs 200 Gb/s per lane, and four lanes, which it calls a station, carry 800 Gb/s[10]. It can address up to 1,024 endpoints, such as accelerators, in one group[10]. AMD's Helios rack, launched in July 2026, joins 72 GPUs and runs UALink over Ethernet for its scale-up network[11].
- **Google's 3D torus.** Google wires 64 of its Ironwood TPU accelerators into each rack, called a cube[24]. Inside the cube each chip links directly to six others, in a three-dimensional grid that Google calls a 3D torus[24]. Optical circuit switches, which steer light from one fiber to another, join cubes into superpods of 9,216 chips and can route around a failed cube or link[24].

Figure 14.2

NVLink per GPU rose from 900 to 3,600 GB/s, Hopper to Rubin

NVLink bandwidth per GPU by generation



Source NVIDIA NVLink product page (NVLink 4 and 5); NVIDIA Rubin launch release, January 2026 (NVLink 6) [23]

Scale-out: InfiniBand versus Ethernet

Scale-out networks typically use InfiniBand or Ethernet, SemiAnalysis wrote in January 2026[20]. InfiniBand is a specialized network long used in supercomputers. But Ethernet switch capacity has grown quickly. Broadcom's Tomahawk 4 switched 25.6 Tbps on one chip in December 2019[25]. Tomahawk 5 reached 51.2 Tbps in August 2022, and Broadcom said it had doubled the bandwidth about every two years since 2014[26]. Tomahawk 6 doubled it again to 102.4 Tbps in June 2025[27]. In June 2025 the Ultra Ethernet Consortium published its 1.0 specification, a set of Ethernet-based rules for communication in AI and supercomputing[28]. It covers how data is carried, how traffic jams are avoided and how one machine reads or writes another's memory directly[28].

10 UALink™ 200G 1.0 Specification Overview. UALink Consortium, 15 May 2025. ualinkconsortium.org

11 AAI 2026: AMD Launches AMD Helios Rackscale Solution for Frontier AI. AMD Newsroom, 23 July 2026. newsroom.amd.com

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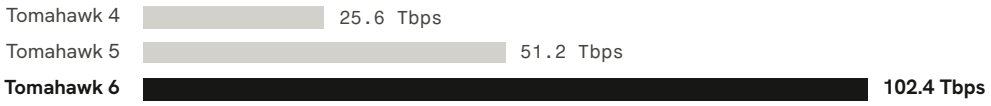
27 Broadcom Ships Tomahawk 6: World's First 102.4 Tbps Switch. Broadcom, 3 June 2025. investors.broadcom.com

28 Ultra Ethernet Consortium (UEC) Launches Specification 1.0 Transforming... Ultra Ethernet Consortium, 11 June 2025. ultraethernet.org

Figure 14.3

Tomahawk switch chip capacity rose from 25.6 Tbps to 102.4 Tbps

Broadcom Tomahawk switching capacity on one chip



Source Broadcom product releases for Tomahawk 4, 5 and 6 [27]

Nvidia sells switches and network cards for both InfiniBand and Ethernet. Its data center revenue, which counts its GPUs as well as its networking gear, reached $89.0 billion in the quarter to July 2026, up 117 percent from a year earlier[29]. In the quarter before, it credited its data center growth to Blackwell 300 products and to demand for its "InfiniBand, Spectrum-X Ethernet, and NVLink" solutions[5].

Optics: pluggable, then co-packaged

Broadcom and Nvidia are moving the optics out of pluggable transceivers and into the switch chip's own package, a design called co-packaged optics. A cluster needs several transceivers, the modules that turn electricity into light and back, for each accelerator. Their power cost is one reason for the change. Broadcom says its Tomahawk 6 Davisson cuts the power used by optical links by 70 percent compared with traditional pluggable modules[30]. It keeps its lasers in field-replaceable modules[30]. Nvidia's Spectrum-X Photonics switches, announced in March 2025, have up to 512 ports of 800 gigabits per second each, and Nvidia says they use four times fewer lasers than traditional designs[31]. Nvidia builds them on its Spectrum-6 switch design[23]. In August 2026 Nvidia said its Spectrum-6 switch systems, with pluggable or co-packaged optics, were arriving at AI data centers around the world[32].

03

Who makes it

Chinese makers of optical modules have built plants outside China. InnoLight has run a plant in Thailand since 2019, alongside its sites in Suzhou and Taiwan[8]. The Thai subsidiary of Eoptolink, another Chinese module maker, lists sites in Chonburi and Rayong[33].

A US rule from July 2026 may reach optical transceivers made by firms on a US security list. The list is the Federal Communications Commission's Covered

5 NVIDIA CORP, Form 10-Q quarterly report for the period ended 2026-04-26 (10-Q). U.S. Securities and Exchange Commission, 20 May 2026. sec.gov

8 InnoLight Technology, company overview page. InnoLight Technology, accessed 9 September 2026. innolight.com

23 NVIDIA Kicks Off the Next Generation of AI With Rubin. Nvidia, 5 January 2026. nvidianews.nvidia.com

27 Broadcom Ships Tomahawk 6: World's First 102.4 Tbps Switch. Broadcom, 3 June 2025. investors.broadcom.com

29 NVIDIA CORP, Form 10-Q quarterly report for the period ended 2026-07-26 (10-Q). U.S. Securities and Exchange Commission, 26 August 2026. sec.gov

30 Broadcom Announces Tomahawk 6 – Davisson, the Industry's First 102.4-Tbps... Broadcom, 8 October 2025. investors.broadcom.com

31 NVIDIA Announces Spectrum-X Photonics, Co-Packaged Optics Networking Switches... Nvidia, 18 March 2025. nvidianews.nvidia.com

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33 Eoptolink Technology, contacts page. Eoptolink Technology, accessed 9 September 2026. eoptolink.com

List of equipment and services judged to pose an unacceptable risk to US national security or to the security and safety of US persons[34]. Under its July 2026 order, the FCC no longer approves new devices for the US market if they contain logic-bearing hardware components made by firms on the list[35]. The order counts as logic-bearing any part that uses digital logic, from a single chip to a whole module, and leaves out parts with no digital function, such as wiring and housings[36]. Commenters in the record named optical transceivers as one such component[37]. As of 16 October 2026, the Covered List did not include InnoLight[38].

China produces most of the world's indium, the raw material for indium phosphide lasers, and restricts its export. Indium phosphide is used in the high-speed laser diodes and photodetectors for optical communications[13]. China accounted for 70 percent of world indium production in 2025[13]. In February 2025 China's Ministry of Commerce placed indium under new export restrictions[13].

AI data centers were straining supplies of the lasers inside optical modules and pushing up their prices, Nikkei Asia reported in May 2026[2]. Nvidia had agreed in March 2026 to invest $2 billion in one laser maker, Lumentum, alongside a multibillion-dollar purchase commitment and rights to future capacity for advanced laser components[3]. Coherent, another laser maker, broke ground in Sherman, Texas, in June 2026 on an expansion of its factory there, which makes optical devices from indium phosphide[15].

Two American firms, Broadcom and Marvell, sell both custom AI chips and Ethernet switch chips. In its annual report for fiscal 2025, Broadcom said its AI chips include Ethernet switch chips as well as custom accelerators, which it builds to each customer's specifications[39]. Its AI semiconductor revenue reached $16.7 billion in the third quarter of fiscal 2026, up 221 percent from a year earlier[6]. Marvell also builds custom AI chips to each customer's specifications and sells Ethernet switch chips, its annual report for fiscal 2026 says[40]. Marvell's five-year agreement with Amazon Web Services, announced in December 2024, covers custom AI chips alongside the optical and networking parts that go with them[41].

2 AI demand strains supplies of lasers, fiber and other optical tech. Nikkei Asia, 27 May 2026. asia.nikkei.com

3 NVIDIA Announces Strategic Partnership With Lumentum to Develop... Nvidia, 2 March 2026. nvidianews.nvidia.com

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34 Protecting Against National Security Threats to the Communications Supply Chain... U.S. Federal Communications Commission, 23 July 2026, p. 3. docs.fcc.gov

35 Protecting Against National Security Threats to the Communications Supply Chain... U.S. Federal Communications Commission, 23 July 2026, p. 6. docs.fcc.gov

36 Protecting Against National Security Threats to the Communications Supply Chain... U.S. Federal Communications Commission, 23 July 2026, p. 13. docs.fcc.gov

37 Protecting Against National Security Threats to the Communications Supply Chain... U.S. Federal Communications Commission, 23 July 2026, p. 10. docs.fcc.gov

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39 Broadcom Inc., Form 10-K annual report for the period ended 2025-11-02 (10-K). U.S. Securities and Exchange Commission, 18 December 2025. sec.gov

40 Marvell Technology, Inc., Form 10-K annual report for the period ended... U.S. Securities and Exchange Commission, 11 March 2026. sec.gov

41 Marvell Technology, Inc., Form 8-K current report for the period ended... U.S. Securities and Exchange Commission, 2 December 2024. sec.gov

Taiwanese firms do much of the physical assembly. Foxconn booked NT$8.1 trillion of revenue in 2025, up 18 percent, and cloud and networking was its largest product category in the fourth quarter[7]. In the same results release, its chairman called Foxconn the world's largest AI server provider[7].

Some assembly is moving to the United States. In April 2025 Nvidia said it was building supercomputer manufacturing plants in Texas, with Foxconn in Houston and Wistron in Dallas[42]. It expected mass production to ramp up in the following 12 to 15 months[42]. In August 2026 Nvidia said Wistron's new plant in Fort Worth, Texas, near Dallas, was producing its GB300 superchips, which pair Nvidia's Grace processor with its Blackwell Ultra GPUs[16]. It said Foxconn was building a factory in Houston to make Nvidia's AI systems[16].

04

The chokepoint

Many firms can build the racks. In October 2025 Nvidia said more than 50 makers of systems and components were gearing up for the Vera Rubin NVL72, which follows MGX, Nvidia's standard design for trays and racks[18]. But in its filing for the quarter to July 2026, Nvidia said the complexity of producing its data center systems had caused production delays[29].

Four inputs to the rack are harder to source elsewhere.

- **Indium phosphide laser capacity.** Nvidia has secured rights to future capacity from laser makers. Alongside its Lumentum deal, in March 2026 Nvidia agreed to invest $2 billion in Coherent[9]. The deal also includes a multibillion-dollar purchase commitment and rights to future access and capacity for Coherent's advanced laser and optical networking products[9].
- **Indium phosphide substrates.** China added indium phosphide substrates, the wafers the lasers are grown on, to its export control list on 4 February 2025[14]. AXT, a US-listed maker of these wafers, does all its manufacturing in China[14]. In its filing for the quarter to June 2026, AXT called the need for Chinese export permits the most significant challenge it faces[14]. It said it could not estimate when it would resume shipping the wafers to the United States[14]. The same filing names Sumitomo and JX, both Japanese firms, as AXT's competitors in indium phosphide[14].
- **NVSwitch.** Nvidia's annual report for fiscal 2026 lists NVLink interconnects and switches among its own networking products[1]. UALink is an open standard designed for accelerator communications within one scale-up group[10]. Since May 2025 Nvidia's NVLink Fusion program has let other

1 NVIDIA CORP, Form 10-K annual report for the period ended 2026-01-25 (10-K). U.S. Securities and Exchange Commission, 25 February 2026. sec.gov

7 Hon Hai Technology Group (Foxconn) Announces FY2025 & 4Q25 Financial Results. Hon Hai Precision Industry (Foxconn), 16 March 2026. honhai.com

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18 NVIDIA, Partners Drive Next-Gen Efficient Gigawatt AI Factories in Buildup for... Nvidia, 13 October 2025. blogs.nvidia.com

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42 NVIDIA to Manufacture American-Made AI Supercomputers in US for First Time. Nvidia, 14 April 2025. blogs.nvidia.com

firms' custom chips use NVLink, including CPUs that Fujitsu and Qualcomm plan to connect to Nvidia GPUs[12].

- **High-layer-count circuit boards.** TTM Technologies, an American circuit board maker, says its circuit board competitors are mostly based in China and Taiwan[43]. Nitto Boseki, a Japanese glass fiber maker, makes NE-glass, a glass cloth that absorbs less signal, for the boards in data center switches and routers[44]. See Substrates and PCBs.

Optical module assembly can move, as InnoLight's high-volume plant in Thailand, opened in 2019, shows[8].

Four questions to answer from memory

1. What is an AI rack? `see How it works`
2. Why do the longer links between AI chips carry signals as light? `see How it works`
3. Which part of an AI rack was scarce in 2026? `see The chokepoint`
4. How much of the world's indium did China produce in 2025? `see The chokepoint`

Four things to remember

— An AI rack wires 72 chips together to run as one computer.

— Almost every link longer than 7 meters carries its signal as light, because copper fades over longer runs.

— Nvidia's NVLink switch chips and the lasers are the hardest parts to replace. In October 2025 more than 50 makers of systems and components were gearing up to build Nvidia's next rack.

— China produced about 70 percent of the world's indium in 2025 and has required export permits for indium phosphide substrates since February 2025.

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A company filings, governments, standards bodies, universities and research institutions; B SemiAnalysis, Reuters, the Financial Times and Nikkei Asia.

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